Slide 1 — DuckDB 101

MSIS-2407 — DBMS - Design, Development & Administration

DuckDB Basics — the database engine for this course

Topics:

- What is DuckDB
- Installation
- Reading CSV files
- Creating tables
- Running SQL queries
- DuckDB + Pandas
- Multiple CSV files
- DuckDB vs Spark / BigQuery

Slide 2 — What is DuckDB?

DuckDB is an **in-process analytical database**.

Key idea:

It runs **inside your application** (Python, R, a Marimo notebook).

Characteristics:

- columnar database
- optimized for analytics
- no server required
- extremely fast for large datasets

Think of DuckDB as:

SQLite for analytics

Slide 3 — Why Use DuckDB in This Course?

Advantages for students:

- No database server setup
- Works directly with CSV files
- Fast analytics queries
- Runs inside **Marimo notebooks** — every lab in this course
- Supports full SQL: keys, constraints, transactions, views

Typical workflow:

CSV → DuckDB → SQL → results as a table

Slide 4 — DuckDB vs Traditional Databases

Feature	DuckDB	MySQL / PostgreSQL
Server required	❌ No	✔ Yes
Installation complexity	Very easy	Medium
Analytics performance	Excellent	Good
Runs inside Python	✔ Yes	Limited

DuckDB is designed for:

analytics, data science, and exploration

Slide 5 — Installing DuckDB

Install using pip:

```
python3 -m pip install "duckdb>=2.0"
```

Verify installation:

```
import duckdb
print(duckdb.__version__)   # 2.x.x
```

Works on Mac, Windows, and Linux.

Full step-by-step setup: the course's `software_installation/` guide.

Slide 6 — Using DuckDB in Python

Example:

```
import duckdb

con = duckdb.connect()

con.execute("SELECT 1+1").fetchall()
```

Output:

```
[(2,)]
```

DuckDB executes **standard SQL queries**.

Slide 7 — Reading a CSV File

DuckDB can read CSV files directly.

```
SELECT *
FROM read_csv_auto('orders.csv')
LIMIT 5;
```

DuckDB automatically:

- detects column names
- detects data types

Slide 8 — Creating a Table from CSV

Convert CSV → table:

```
CREATE OR REPLACE TABLE orders AS
SELECT *
FROM read_csv_auto('orders.csv');
```

Now the table **orders** exists in DuckDB.

`CREATE OR REPLACE` lets you re-run the cell without an error.

Slide 9 — Query Example

```sql
SELECT customer_id,
       COUNT(order_id) AS num_orders
FROM orders
GROUP BY customer_id
ORDER BY num_orders DESC
LIMIT 10;
```

Business question:

Which customers placed the most orders?

Slide 10 — DuckDB + Pandas

Using the same connection `con` that created `orders`:

```
df = con.execute("""
    SELECT *
    FROM orders
    LIMIT 10
""").df()

df
```

DuckDB returns a **Pandas DataFrame** — Marimo shows it as a table.

Slide 11 — Query a CSV Without Loading It

```
df = con.execute("""
    SELECT category, COUNT(*) AS num_sales
    FROM read_csv_auto('sales.csv')
    GROUP BY category
    ORDER BY num_sales DESC
""").df()
```

No `CREATE TABLE` needed — DuckDB reads the file on the fly.

Workflow:

CSV → SQL → DataFrame

Slide 12 — Reading Multiple CSV Files

```
SELECT *
FROM read_csv_auto('data/orders_*.csv');
```

DuckDB automatically loads all matching files.

Useful for:

- partitioned datasets
- large data collections

Slide 13 — DuckDB vs Spark vs BigQuery

System	Best Use Case
DuckDB	Local analytics
Spark	Distributed big data
BigQuery	Cloud warehouse analytics

Rule of thumb:

```
DuckDB   → laptop
Spark    → cluster
BigQuery → cloud
```

Slide 14 — DuckDB in 30 Seconds

Complete workflow example:

```
import duckdb

con = duckdb.connect()

df = con.execute("""
    SELECT category, SUM(sales_amount) AS revenue
    FROM read_csv_auto('sales.csv')
    GROUP BY category
    ORDER BY revenue DESC
""").df()

df
```

Pipeline:

CSV → DuckDB → SQL → Pandas DataFrame

Slide 15 — Summary

DuckDB is:

- lightweight
- fast
- easy to install
- ideal for analytics — and for learning database design

Typical workflow:

CSV → DuckDB → SQL → DataFrame

Slide 16 — DuckDB in MSIS-2407

We will use DuckDB throughout this course to:

- build schemas with keys and constraints
- load and validate data
- run transactions and see concurrency at work
- build and query star schemas
- practice security, backup, and data quality

Every lab is a Marimo notebook running on DuckDB.